AI-BASED SMART FOOD WASTE & DEMAND PREDICTION SYSTEM

VITYARTHI FLIPPED COURSE

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Food waste is a common problem in hostels and college mess systems. Many times, food is either prepared in excess or falls short because there is no proper way to estimate how many students will eat on a particular day.

This project presents a simple AI-based system that predicts the number of students expected in the mess using past data like previous attendance, day of the week, and whether any event is happening. Based on this prediction, the system suggests how much food should be prepared.

The main aim of this project is to reduce food wastage and improve planning using basic machine learning and logical rules.

INTRODUCTION

In most hostel mess systems, food preparation is done based on guesswork. This often leads to unnecessary wastage or shortage of food. Managing food efficiently is important not only to save resources but also to ensure that students are satisfied.

With the help of Artificial Intelligence and Machine Learning, we can use past data to make better decisions. This project focuses on building a simple system that predicts daily attendance in a mess and gives suggestions accordingly.

The system is easy to use and shows how even basic AI techniques can solve real-life problems.

PROBLEM DEFINITION

Mess management systems usually do not have a proper method to estimate the number of students who will eat food each day. Because of this:

Extra food gets wasted

Sometimes food is not enough

Resources like money and effort are not used properly

So, there is a need for a system that can predict attendance and help in better planning of food preparation.

OBJECTIVES

The main goals of this project are:

To predict the number of students coming to the mess

To calculate how much food should be prepared

To reduce food wastage

To use basic machine learning in a real-life problem

To give simple suggestions for better decision making

REQUIREMENT ANALYSIS

- Functional Requirements

User should be able to enter:

Day of the week

Previous attendance

Event status

System should predict expected attendance

System should suggest food quantity

System should give a basic recommendation

- Non-Functional Requirements

System should be easy to use

Output should be quick

Code should be simple and understandable

Should work on a normal computer

- Hardware Requirements

Laptop or PC

Minimum 4GB RAM

- Software Requirements

Python

Required libraries:

pandas

numpy

scikit-learn

Feasibility

Easy to implement

Low cost